Tier-D Rank 22: Backbone Closure, Soundness Gate, and Vivify Handoff

Tactix engineering handoff

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Objective and outcome

This continuation resumed the frozen Tier-D ranks 21–40 convergence round without duplicating the six already checked wins: ranks 21, 23, 25, 28, 29, and 38. Rank 22 remains the first open target. The immediate objective was to finish the reopened canonical Backbone parity item under the development plan's N1–N4 constraints, then identify the first downstream owner if Rank 22 still diverged.

The outcome is a mechanism-level closure of `SP-BACKBONE-3`. At the common conflict-4487 Probe phase, Tactix now matches Kissat's complete early Backbone computation: candidate schedule, every derived unit, every binary closure, every retained round, private effort accounting, the budget stop, and the post-round root closures. A separate rule-boundary trace proves that both solvers leave this Backbone call with the same semantic population of 3,442,398 variables. Vivify is the first later owner: it exits with 3,442,395 unassigned-active variables in Tactix and 3,442,270 active variables in Kissat. The later Backbone computation therefore inherits its population mismatch and must not be reopened for that symptom.

Implementation decisions

The repair follows Kissat's chained binary-UIP batch rather than tuning a budget or adding a witness-specific gate.

- Chained binary implications are assigned at the current accumulated assumption level. A derived unit remains speculative through the rest of its round; the complete round suffix is unwound by trail position before the unit batch is replayed at root.
- Later computations exclude variables already fixed at root, matching Kissat's `ACTIVE_VAR` representation while retaining Tactix's public variable identities and complete one-literal `backbone_try` surface.
- The binary watch cursor consumes the first fetched large-watch word on an early stop. Conversely, a conflicting binary returns before charging that literal's scan or private ticks, matching the reference control flow exactly.
- Post-round general root propagation is outside Backbone's private meter. Startup propagation, both Lucky trials, sparse root collection, and initial Probe are likewise excluded from `search_ticks`; initial Probe leaves the search-time Probe anchor at zero as in Kissat.

- Binary-only propagation records the trail prefix whose large watches it skipped. Every full propagation entry settles that debt, and verdicts and guards use the resulting common fixpoint predicate.

These changes are documented in `docs/calculus.md`, `docs/implementation-notes.md`, and the closed `SP-BACKBONE-3` row of `docs/scheduling-parity-ledger.md`. Focused regressions cover the speculative-unit lifecycle, current-level propagation, root replay and its meter boundary, the first large watch word, the uncharged binary-conflict scan, large-clause root closure, and native-LRAT output.

Exact evidence

All real-instance work used immutable pins on Slurm with two CPUs, 16 GiB, no node constraint, and no array throttle.

Job 1636147. All 67 derived units match in external literal, binary closure, scan count, word count, and tick count. Its aggregate residual isolated the reference's uncharged conflicting-binary scan.

Job 1636165. After the cursor correction, all 23 computation-2 rounds match in retained candidate count and hash, units, probes, scans, words, ticks, and the 40,154,973-tick budget stop.

Job 1636177. Filtering root-fixed variables makes the computation-2 initial schedule exact at 6,884,986 candidates; every retained round remains exact.

Jobs 1636196 and 1636219. The eight post-round root-closure counts are 5, 19, 8, 4, 4, 47, 4, and 4 on both implementations. The corrected Kissat observer reads the authoritative active-variable delta after its flushed trail, while Tactix reads the retained root trail.

Jobs 1636337 and 1636338. Rule-boundary snapshots preserve the common conflict, decision count, and pre-Probe formula. The early Backbone exits at 3,442,398 semantic variables on both sides. Vivify then makes the first population split. These jobs are diagnostic overlays, not replacements for the frozen performance authority, and were cancelled once the boundary was captured.

The primary campaign artifacts are under `benchmarks/runs/20260820-0550-tierd-rank22-backbone-unit-accounting/` through `benchmarks/runs/20260820-0633-tierd-rank22-backbone-root-implied-reference/`, plus the paired `benchmarks/runs/20260820-0705-tierd-rank22-rule-boundary-*` directories.

Mandatory soundness escalation

The first quick random-policy corpus replay, Slurm job `1636407`, found two real failures: corpus 004 returned a SAT model falsifying clause 199, and corpus 023 returned an UNSAT proof rejected by `lrat-check`. This fired the development plan's mandatory Claude escalation. The first static repair passed focused tests but not either exact case in Slurm job `1636634`; it was therefore rejected rather than promoted.

Claude then reproduced both cases with their recorded schedules and real checkers. Their common cause was a sessionless Sweep substitution: an early return left the transaction's retiring-literal

list empty, so the physical rewrite could delete one direction of a binary while retaining the other. That half-watched clause could lose its proof identity and poison LRAT, or a whole binary could disappear and permit a wrong model. The repaired session records membership independently of the optional dense ledger, with a focused mirror-and-identity regression and a kernel invariant for unnamed binary reasons.

Validation fuzzing then found corpus 026: Sweep's RUP admission replay was a no-op outside native LRAT, so DRAT and proofless runs could admit lemmas that unit propagation could not rederive. An explicit `RupReplay` purpose now separates Congruence's chain-only use from the format-independent admission decision shared by Sweep and HyperBinaryResolution. The minimized DRAT case and a proof-format parity regression pin the repair; the complete diagnosis is `benchmarks/reports/2026-08-20-sweep-drat-admission-gate.md`.

The final exact model, native-LRAT, and DRAT reproducers pass. Four complete 26-entry corpus configurations report zero failures. Release passes 72/72, Trace 71/71, ASAN-O3 72/72, and Debug 71/71 tests; Debug's multi-hour scripted demo is covered by the other three configurations. Additional random-policy fuzzing covers 1,200 DRAT and 800 native-LRAT cases with zero failures and direct Sweep and HyperBinaryResolution reachability. Final Slurm job `1636794` exercises the immutable Debug pin `8662f1f1...` on 10,000 more random-policy cases: zero failures, 8,771 model/proof/differential-checked verdicts, and direct Backbone, Sweep, and HyperBinaryResolution reachability. Leak detection remains disabled only for the known LeakSanitizer-under-ptrace environment limitation.

Limitations and next actions

Rank 22 is not claimed solved. Backbone is closed only at a common-input rule boundary; the first remaining mismatch is inside Vivify. The next diagnosis must compare Vivify's schedule, per-band budget, candidate terminal actions, and root-unit yield from the common 3,442,398-variable input. Existing closed Vivify rows should be reopened only if that trace violates their stated conditions; no constant, rank, fingerprint, or deadline tuning is admissible.

Commit and push this coherent Backbone and soundness tranche, then run the checked 28B+5C guard and rank-22 16 GiB/1,200-second witness from an immutable production pin. The next performance diagnosis starts inside common-input Vivify; the soundness fixes do not authorize reopening Backbone or tuning a witness. The active worktree must remain until the continuing Tier-D branch is integrated; obsolete diagnostic overlays may be removed once no running job references them.